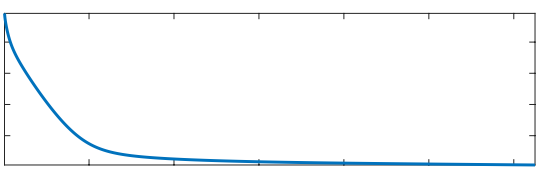


$f(x_k)$



$\log_{10}(f(x_k) - f(x^\star))$